Incentive-Aware Multi-Objective Matching for Peer-to-Peer Commuter Carpooling

Krunal Savaj [1]

Abstract

1. Introduction

Urban commuting accounts for a significant share of daily vehicle miles traveled in the United States, yet the average private car carries only 1.08 occupants during peak hours (Santos et al., 2011). The math is simple: if even a fraction of those solo commuters shared rides with someone heading the same direction, the impact on congestion, cost, and emissions would be substantial — and it requires zero new infrastructure.

Platforms like Uber and Lyft already offer pooled rides, but their matching algorithms are built around professional drivers who accept any profitable trip. Peer-to-peer (P2P) carpooling between regular commuters is a fundamentally different problem. These are people with fixed routes, fixed schedules, and limited tolerance for detours. A commuter who drives from the Bronx to Midtown every morning at 8:15 AM is not going to accept a 6 km detour to pick up a stranger unless the economics make sense.

The Learning Problem. We frame this as a supervised prediction task followed by combinatorial optimization. Given a set of n drivers $\mathcal{D} = \{d_1, \ldots, d_n\}$ and m riders $\mathcal{R} = \{r_1, \ldots, r_m\}$, each defined by an origin–destination coordinate pair, pickup time, and fare, we construct a candidate pair (d_i, r_j) whenever the geodesic distance between their origins falls within a configurable spatial threshold δ_{max}.

For each candidate pair, we engineer a feature vector $\mathbf{x}_{ij} \in \mathbb{R}^6$ capturing route overlap, detour cost, temporal compatibility, and economic incentives. The prediction task has two objectives:

1. **Regression:** Predict the additional detour $\hat{y}_{ij}^{\text{det}} \in \mathbb{R}^+$ the driver would incur.

2. **Classification:** Predict the probability $\hat{y}_{ij}^{\text{acc}} \in [0, 1]$ that the driver accepts the match.

These dual predictions are fused into a composite match score $s_{ij} = \hat{y}_{ij}^{\text{acc}} - \lambda \cdot \hat{y}_{ij}^{\text{det}}$, where λ controls the detour penalty. At inference time, the scores serve as edge weights in a bipartite graph $G = (\mathcal{D}, \mathcal{R}, E)$, and we solve for the maximum-weight matching to produce globally optimal one-to-one assignments.

Why This is Hard. This problem is hard for a few reasons. No public P2P carpooling dataset exists — real matching data is proprietary (Uber, Waze), so we have to construct a simulation pipeline that synthesizes labeled training pairs from 3.5 million single-passenger NYC taxi trips. The acceptance decision itself is inherently multi-factor: a driver's willingness depends on the interaction between detour severity, fare compensation, timing, and the detour-to-trip-length ratio, not any single threshold. And even with good pairwise predictions, greedy assignment does not guarantee a good global outcome — a locally attractive match can block a better pairing downstream.

Our Approach. Our pipeline has three stages: (1) a data synthesis engine that generates geographically realistic driver–rider pairs from public taxi records using corridor-aware coordinate projection and multi-factor stochastic labeling, (2) a dual-objective XGBoost model (Chen & Guestrin, 2016) that simultaneously predicts detour cost and acceptance probability, and (3) a bipartite matching layer (Kuhn, 1955) that selects globally optimal assignments using the predicted scores as edge weights. This architecture follows the learning-augmented matching paradigm (Bjorck et al., 2021), which provably outperforms either pure heuristic matching or pure ML prediction alone.

2. Related Work

Ride-sharing optimization has been studied from several angles — shareability analysis, matching algorithms,

[1]Khoury College of Computer Sciences, Northeastern University, Portland, ME. Correspondence to: Krunal Savaj <savaj.k@northeastern.edu>.

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trajectory-based routing, and multi-objective formulations. We review each below.

Shareability Networks and Fleet Optimization. The foundational work of Santi et al. (2014) introduced *shareability networks*, modeling every NYC taxi trip as a node and connecting shareable trips with edges. By solving bipartite matching on this graph, they showed that ride-sharing could reduce the taxi fleet by 40% with minimal passenger delay. Their approach relies entirely on fixed spatiotemporal thresholds to determine shareability — if two trips can be combined within a delay budget, they are connected. While powerful for fleet-level analysis, this binary threshold ignores the nuance of *individual* willingness: a 3 km detour is acceptable on a 30 km highway commute but painful on a 5 km city trip. Our work builds directly on their graph formulation but replaces the fixed threshold with a learned acceptance model that captures these non-linear trade-offs.

Matching with Learned Predictions. Bjorck et al. (2021) formalized the concept of *learning-augmented matching*, proving that using ML predictions as edge weights in bipartite optimization provably outperforms both pure heuristic matching and pure ML prediction used in isolation. Their theoretical guarantees show that even when predictions are imperfect, the matching quality degrades gracefully (never worse than $2\times$ optimal). They validate on taxi-rider matching with uncertain trip duration. Our pipeline directly instantiates this paradigm: XGBoost generates match scores, and maximum-weight bipartite matching (Kuhn, 1955) selects globally optimal assignments at inference time.

Spatial and Temporal Pre-filtering. Efficient candidate generation is critical at scale. Brute-force comparison of all driver–rider pairs is $O(n^2)$ and infeasible for large networks. Anagnostopoulos & Ramson (2025) address this with a three-dimensional matching framework incorporating spatial, temporal, and social behavior similarity, tested on Athens taxi data. Their social dimension — modeling whether passengers have compatible travel habits — is particularly relevant for P2P settings with repeat commuters. We adopt a similar spatial and temporal pre-filtering strategy, using BallTree indexing with the Haversine metric (Omohundro, 1989) for $O(\log n)$ neighbor queries within configurable geodesic radii, combined with rolling temporal batch windows.

Trajectory-Based Routing. Hsieh (2017) proposed matching based on full GPS trajectories rather than origin–destination points, integrating Google Maps API for real road-network distances. This captures critical routing details that straight-line (geodesic) distance misses: a rider 3 km away across a river with no bridge requires a 20-minute detour, while one 3 km down the highway requires 30 seconds. Our current implementation uses geodesic distance via `geopy` (GeoPy Contributors, 2023), which is a known limitation. However, for commuter corridors where routes are predominantly highway-based, the geodesic approximation is reasonable, and we discuss OSRM integration as future work.

Gradient Boosted Trees for Tabular Geospatial Data. For structured, tabular prediction tasks, XGBoost (Chen & Guestrin, 2016) tends to outperform deep learning when the feature space has moderate dimensionality and mixed types (Grinsztajn et al., 2022). Several recent studies have applied XGBoost specifically to ride-sharing contexts, including dynamic rider grouping in NYC based on route and time similarity (Ma & Li, 2023). These works validate that gradient-boosted trees are well-suited to our feature space (distances, ratios, fares, timestamps) and motivate our choice of XGBoost over neural architectures.

Multi-Objective Optimization Approaches. An alternative paradigm treats ride-matching as a multi-objective optimization problem. Wang et al. (2023) apply NSGA-II (a Pareto-based evolutionary algorithm) to minimize both driver detour time and passenger travel time subject to time-window and vehicle capacity constraints. While evolutionary methods can explore the Pareto frontier directly, they are computationally expensive and do not produce a trained model that generalizes to unseen pairs. Our approach separates prediction (learning the detour–acceptance relationship) from optimization (bipartite matching), which allows the predictive model to generalize across regions and the matching layer to be swapped independently.

How Our Work Differs. While shareability networks (Santi et al., 2014) and learning-augmented matching (Bjorck et al., 2021) have separately addressed ride-pooling optimization, no prior work combines all of the following: (1) a reproducible data synthesis pipeline that generates labeled P2P training pairs from public single-passenger taxi records using corridor-aware coordinate projection, (2) a dual-objective gradient-boosted model that simultaneously predicts detour cost and acceptance probability, (3) multi-factor stochastic labeling with Gumbel noise to simulate realistic human decision-making, and (4) a bipartite matching layer that uses the predicted scores for globally optimal assignments. As a baseline, we use logistic regression (Cox, 1958) via scikit-learn (Pedregosa et al., 2011), which cannot capture these interaction effects, motivating the non-linear XGBoost model. Our features are dimensionless ratios (e.g., overlap ratio) rather than raw coordinates, so the trained model transfers across cities without retraining.

3. Data

4. Methods

4.1. Evaluation Metrics

4.2. Baseline Model

4.3. Advanced Model

4.4. Training Procedure

5. Results

6. Discussion

6.1. Limitations

6.2. Ethical Considerations

7. Conclusion

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